FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY

BACHELOR IN INFORMATION TECHNOLOGY (HONOURS)
BACHELOR IN COMPUTER SCIENCE (HONOURS)
(ODL AND WEEKEND PROGRAMME)

BIT1144/BTL114/BCL1144 PROGRAMMING FUNDAMENTALS

INDIVIDUAL ASSESSMENT

FINAL ASSESSMENT 40%:
Integrated System Development Project

DUE DATE SUBMISSION: 05 APRIL 2026, 11:59PM

STUDENT NAME	
STUDENT ID	

DECLARATION OF INTEGRITY:

I hereby declare that this lab assessment submission is my own independent work and does not contain plagiarized content, unauthorized assistance, or any form of academic dishonesty. I confirm that I have adhered to the academic integrity policies outlined by the university.

I understand that if any form of academic dishonesty, including plagiarism, falsification of results, or unauthorized collaboration, is detected in this submission, I may face disciplinary actions. This may include, but is not limited to, receiving a failing grade for this assessment or further academic penalties as determined by the university's academic integrity committee.

By signing below, I acknowledge my understanding and agreement with this declaration.

Student's Signature: _____ Date: _____

ASSESSMENT OVERVIEW

This Final Project requires students to **extend and enhance** the system developed in Assignment 2 into a complete, professionally structured Java application.

Unlike Assignment 2 (prototype development), this assessment emphasizes:

- System refinement
- Professional responsibility
- Ethical programming practices
- Documentation quality
- Reflection and improvement

The project measures both technical integration (CLO2) and professional practice (CLO3), with stronger emphasis on CLO3.

CLO–PLO ALIGNMENT

CLO	Description	Bloom Level	PLO	Weight
CLO2	Develop Java-based applications using appropriate programming principles	C3–P3	PLO3	60%
CLO3	Demonstrate responsible and ethical programming practices	A3	PLO6	40%

ASSESSMENT OBJECTIVES

This assessment evaluates students' ability to:

- Apply OOP principles in structured problem solving
- Implement file handling operations
- Develop a functional GUI
- Demonstrate professional coding practices

TABLE OF SPECIFICATION (ToS)

Component	CLO	Marks	%
System Architecture & Design	CLO2	20	20%
Functional Completeness	CLO2	20	20%
GUI & User Interaction	CLO2	10	10%

Validation & Exception Handling	CLO2	10	10%
Documentation Quality	CLO3	15	15%
Reflection & Professional Insight	CLO3	15	15%
Ethical Practice & Integrity	CLO3	10	10%
Total		**100**	**100%**

SLT JUSTIFICATION (28 HOURS)

Activity	**Hours**
System redesign	6
Feature expansion	14
GUI improvement	6
Validation & debugging	6
Documentation	4
Reflection writing	4
Total	**40 Hours**

Submission Instruction

This assignment must be submitted via the official Learning Management System (LMS) before the stated deadline. Email submissions will NOT be accepted unless officially approved due to documented technical issues.

Required Submission Files

Students must upload a single compressed file (.zip) containing:

A. Source Code

- All .java files (Ensure code compiles and runs)

B. Executable (If Applicable)

- .jar file OR instructions to run program

C. Documentation Report (PDF format)

D. Screenshot Evidence

File Naming Convention

The submission file must follow this format: CourseCode_Assignment2_StudentID_Name.zip

Example:

OOP101_Assignment2_20234567_AhmadAli.zip

Improper file naming may result in mark deduction.

Technical Requirements Before Submission

Students must ensure:

- The program compiles without error
- The program runs successfully
- All required features are implemented
- No hardcoded file paths (use relative paths)
- No external libraries unless approved

If the program does not run, marks will be deducted accordingly.

AI Usage Policy (If Applicable)

If AI tools (e.g., ChatGPT, Copilot) are used:

- Students must clearly state how the tool was used
- AI cannot replace original coding work
- Students must understand and explain submitted code

Failure to declare AI usage may be considered misconduct.

TASK REQUIREMENTS

Students must continue from the same system selected in Assignment 2. The existing prototype must be improved and expanded.

PROHECT REQUIREMENTS

System Enhancement (Technical – CLO2)

Students must:

- Expand to minimum 5 classes
- Improve class relationships and structure
- Implement full CRUD operations
- Improve data persistence (structured file format)
- Implement comprehensive input validation
- Improve exception handling
- Implement multi-screen GUI OR tab-based interface

Professional & Ethical Requirements (CLO3 – Strong Focus)

Students must include:

1. Full documentation (4–6 pages):
    - System overview
    - Improved class diagram
    - Explanation of design decisions
    - Code improvement from Assignment 2
2. Reflection report (minimum 800 words):
    - Challenges faced
    - Ethical considerations
    - Code improvement discussion
    - Learning reflection
3. Integrity Declaration:
    - AI usage disclosure
    - Code originality statement
4. Version Comparison:
    - Brief comparison between Assignment 2 and Final Project improvements

Marking Scheme

Criteria	CLO	Extended Abstract	Relational	Multi-structural	Uni-structural	Pre-structural	Weight	Mark Allocation
System Architecture	CLO2	18–20 Integrated, scalable design	14–17 Logical integration	10–13 Multiple but loosely linked	5–9 Minimal structure	0–4 Poor design	20	
Functional Completeness	CLO2	18–20 Full, robust CRUD	14–17 All required working	10–13 Most working	5–9 Partial	0–4 Fails	20	
GUI Interaction	CLO2	9–10 Intuitive multi-interface	7–8 Functional UI	5–6 Basic UI	2–4 Minimal	0–1 None	10	
Validation & Exception	CLO2	9–10 Robust validation & recovery	7–8 Proper handling	5–6 Basic handling	2–4 Limited	0–1 None	10	
Documentation Quality	CLO3	14–15 Clear, structured, professional	11–13 Good explanation	8–10 Basic report	4–7 Minimal	0–3 None	15	
Reflection & Insight	CLO3	14–15 Deep insight, ethical awareness	11–13 Good reflection	8–10 Basic reflection	4–7 Limited	0–3 None	15	
Ethical Practice	CLO3	9–10 Clear integrity & AI declaration	7–8 Proper declaration	5–6 Minimal	2–4 Incomplete	0–1 Integrity issue	10	
Total							**100**	

CLO ALIGNMENT SUMMARY

CLO	Total Marks	Percentage
CLO2	90	90%
CLO3	10	10%

PERFORMANCE INTERPRETATION SCALE

Mark Range	Performance
80–100	Excellent – Integrated technical and professional competence
65–79	Good – Strong technical with adequate professionalism
50–64	Satisfactory – Functional with limited reflection
40–49	Weak – Incomplete integration
<40	Unsatisfactory

Overall Feedback

First Marker:

_____
_____
_____
_____

Total Mark: _____

Signature	Date

Second Marker:

_____
_____
_____
_____

Total Mark: _____

Signature	Date